Archives of Computational Methods in Engineering (2023) 30:727–747
https://doi.org/10.1007/s11831-022-09815-7

REVIEW ARTICLE

A Review on Kalman Filter Models

Masoud Khodarahmi[1] · **Vafa Maihami**[1]

Received: 18 December 2021 / Accepted: 26 August 2022 / Published online: 1 October 2022

Abstract

Kalman Filter (KF) that is also known as linear quadratic estimation filter estimates current states of a system through time as recursive using input measurements in mathematical process model. Thus algorithm is implemented in two steps: in the prediction step an estimation of current state of variables in uncertainty conditions is presented. In the next step, after obtaining the measurement, previous estimation is updated by weighted arithmetic mean. Accordingly, using KF in non-linear systems can be difficult. For nonlinear systems Extended KF (EKF) and Unscented KF (UKF) represent the first-order and higher order linear approximations. KF cannot predict appropriate values for modeling system behavior in more complicated systems. In the current study, in addition to referring to basic methods, a review on recent researches on Multiple Model (MM) filters has been done. More reliable estimations obtain by using two or more filters with different models in parallel, by allocating an estimation to each filter, outputs of each filter are calculated. MM Adaptive Estimation (MMAE) and Interacting MM (IMM) are the most used methods for estimating MMs.

1 Introduction

Kalman Filter, KF [1] that is also known as Linear Quadratic Estimation predicts future state of a system based on previous and current states. KF is stated by an equation, but it is separated into two steps including prediction and update. In prediction step, an estimation is obtained for the current state by utilizing a series of state estimations in previous intervals. This predicted estimation is prior knowledge because it is related to previous estimations and no observations exist for the system in the current state. In update step, prior estimation is mixed to the current observations to present an estimation of the current and the next states of the system. These steps are usually repeated alternatively, that is prediction is done until next observation, then update is done using the current observations. If there is no observation in an interval, several update predictions will be performed until the next observation. Similarly if several independent observations are done in an interval, several updates with different H_k matrixes will be obtained based on each observation [2, 3].

✉ Vafa Maihami
Maihami@iausdj.ac.ir

Masoud Khodarahmi
Masoodkhodarahmi@gmail.com

1 Department of Computer Engineering, Islamic Azad University, Sanandaj Branch, Sanandaj, Iran

1.1 Kalman Filter: Application

KF can be used for estimating system parameters, its goal is minimizing error in noise during estimation that minimizes error covariance in special conditions [4]. This filter overcomes deficiencies of Wiener Filter and theory. It is appropriate for estimation of state changes in various systems with multivariable time. It has created the basis for developing modern control theory and signal processing in real time [5]. These days, KF has developed from optimal state estimation to automation, positioning, target tracking, communication and signal processing, digital image processing, speech signal processing, earthquake prediction and many other grounds in engineering. Accordingly, it has been one of most common tools in information, control, and process automation [6]. KF has many usages in science and technology such as tracking, vehicle monitoring especially planes and spacecraft. KF represents wide concepts about time series, signal processing and econometrics. This filter is considered as one of basic concepts in planning and robots monitoring and neural system modeling. Based on time delay between sending commands and receiving replies using KF makes estimation of different states of a system possible [7].

1.2 Kalman Filter: Progress

KFs are ideal for systems that are always changing. Their advantage is that they need a little memory because they do not need memory except for saving information about previous states. These filters are fast therefore they are suitable for real time issues and embedded systems.

Although this filter is utilized in systems with linear dynamic to estimate optimal state, several developments have been achieved during a time period for improving performance condition. Simple method has low effect for nonlinear systems, Extended KF (EKF) can have better results.

When a dynamic system becomes more complicated, EKF may not be convergent or its implementation is very difficult. Furthermore, linearization step includes calculating partial derivatives in each step of repeating filter, these factors make EKF on algorithm with high computational cost. Unscented Kalman Filter (UKF) has been created as an appropriate replacement to solve these problems and improve EKF performance. Since KF observer may not be resistant to uncertainty of the model, Robust KF (RKF) was designed for discrete time systems. State estimation in high dimensions was a basic problem, but by presenting Cubature KF (CKF) a systematic solution was presented for nonlinear filter problems. In some practical usages, system models cannot be obtained accurately. Providing uncertainty about model, filter performance may reduce [8]. In complicated systems behavior modeling is not easy thus Multiple Model Filters (each with a different system) have been designed that can work in parallel afterwards outputs of each filter help to get more reliable estimations. These models have problem in changing model during movement, for solving this problem Interacting Multiple Model (MM) has been presented, that is able to change from one model to another accurately.

For the current study, 45 articles were reviewed. Basic articles date back to 1966 but research has been done on papers that represent multiple methods in 2019–2020.

The current study gives a summary on design and performance of some basic models and aimed at investigating some MMs such as MMAE and IMM that have been presented based on basic models.

2 Purposes and Motivations

The main purpose of the current study is presenting a progress status and developing systems of designed models for estimating system state. For achieving this goal, two subjects are dealt with: (A) Describing main extended methods by previous authors for solving problems and optimization of current filters.

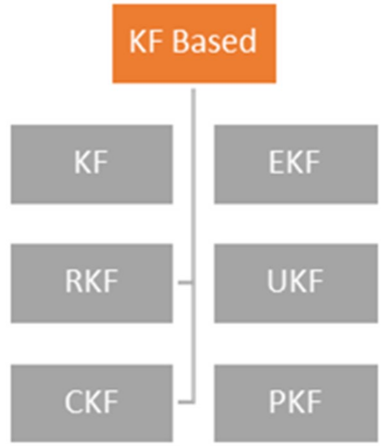


Fig. 1 Basic linear and nonlinear Kalman Filters

(B) Analyzing and discussing the results for understanding what main achievements have been gained before, which problems have not been solved yet that can limit applications, and what research effort is being made to progress.

3 Type of Basic Kalman Filters

Linear transformation is base of KF, but more complicated systems can be non-linear. Being non-linear can occur in observations, modeling or both. In this section, we will review the basic linear and nonlinear models (Fig. 1).

Linear base Kalman filter (Basic Kalman filter)

KF is an optimal estimator that presents a recursive solution for analyzing time series. These unknown states calculate a set of *xi* from a set of measurements of z_i process.

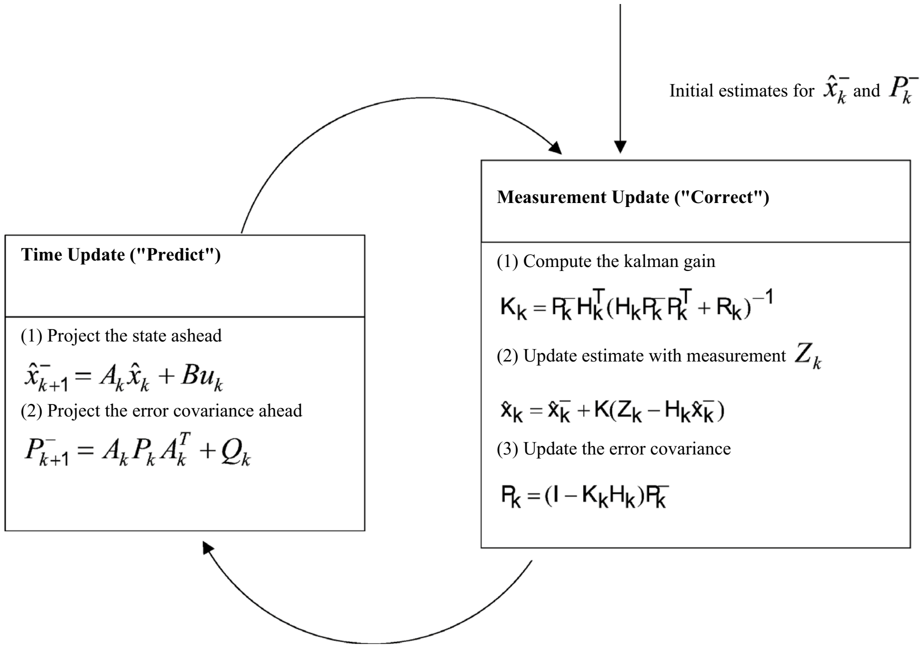
$$\begin{aligned} x_k &= F_{k-1}x_{k-1} + W_{k-1}, \\ z_k &= H_k x_k + V_k. \end{aligned} \tag{1}$$

In which F is state transition, H is observation model, W and V are defined as white noise with covariance QK and RK respectively. Based on equations in (1) KF estimates x_k state based on noisy measurement of z_k. In the equation x_k $z_k = H_k\ x_k + v_k$ is the estimation of the previous state. The previous estimation (that is called prediction in KF) in time k, x_k is state estimation, when all measurements are available until $k-1$, previous state and error covariance are calculated as follow.

$$\begin{aligned} x_k^- &= F_{k-1}x_{k-1}, \\ P_k^- &= F_{k-1}P_{k-1}^+F_{k-1}^T + Q_k. \end{aligned} \tag{2}$$

Next state and error covariance are as follow:

Fig. 2 A complete picture of the operation of the Kalman filter [3]



$$\hat{x}_k = x_k^- + K_k(y_k - H_k x_k^-),$$
$$\hat{P}_k = (I - K_k H_k)P_k^-. \tag{3}$$

After predicting the state and covariance using values of previous estimations, Kalman optimal result is calculated as follow:

$$K_k = P_k^- H_K^T (H_k P_k^- H_K^T + R_k)^{-1}. \tag{4}$$

Then state prediction is updated using Kalman optimal result:

$$\widehat{\mathrm{x}}_{\mathrm{k}} = \widehat{\mathrm{x}}_{\mathrm{k+}}\mathrm{k}_{\mathrm{k}}\left(\mathrm{z}_{\mathrm{k}} - \mathrm{H}_{\mathrm{k}}\widehat{\mathrm{x}}_{\mathrm{k}}\right). \tag{5}$$

Finally, covariance matrix is updated as follow: I is identity matrix (Fig. 2).

$$P_k = (I - K_k H_k)P_k. \tag{6}$$

3.1 Nonlinear Base Kalman Filters

3.1.1 Extended Kalman Filter (EKF): 1966

EKF is the nonlinear version of the KF. In each step of estimation of system states based on estimated data, nonlinear system dynamic is linearized as separable functions using Taylor series. This extended algorithm is done in two steps: in the first step that is called prediction step an initial estimation of state variables in sample k until $k-1$ state is done based on system information. In the second step that is called update step, prediction in the first step is improved based on new available data to obtain final estimation states in sample k [3].

3.1.2 Robust Kalman Filter (RKF): 1994

RKF that has basic structure of a standard KF was designed for unknown discrete time systems with parameter uncertainty that has been able to solve the problem of parameters uncertainty in state and output matrixes, and necessary changes for parameter uncertainty are considered and a solution is obtained for upper limit of error covariance for linear and nonlinear systems by using Riccati equation (methods based on linear matrix inequality). In RKF is also tried to solve parameter uncertainty in state and output matrixes by presenting upper limit for filter error variance [9].

3.1.3 Unscented Kalman Filter (UKF):1997

When a dynamic system becomes more complicated, EFK may not become convergent and its implementation will be very difficult. In addition, linearization includes calculating partial derivatives in each step of filter. These factors change EKF to an algorithm with high computational cost. Reference [10] argued that need to continuous prediction of new state and system observation is the basic problem in using Kalman Filter in nonlinear systems, and introduced UKF that is called Unscented Kalman Filter. This algorithm has two advantages than EKF. It can predict system state more accurately and EKF implementation is very difficult. UKF is considered as a progress than EKF [10]. UKF is based on this principle that estimation of probability distribution is easier than estimation of a function or a random nonlinear transformation. Due to easy calculation process and better performance in nonlinear systems, UKF is a promising method for estimating states of nonlinear dynamic systems, but when system model contains uncertainty, its solution will be destructive or divergent [11].

When states and observations transformation, that is prediction and update are completely nonlinear, EKF will have low efficiency, because covariance is increased in linearization of nonlinear models. UKF is used definitive sampling that is known as Unscented Transform to collect a set of points around the mean, then these points are entered in nonlinear function to obtain new mean and covariance. For definitive systems the amount of mean and covariance is presented by more certainty [12]. This method has been known as Monte Carlo method or Taylor series for previous statistics. In fact, by this method we do not need to calculate basic steps for implementing UKF directly.

Basic steps to implement UKF are as follow:

(1) Calculating set of sigma points.
(2) Allocating weights to each sigma points.
(3) Transforming sigma points using nonlinear function.
(4) Calculating Gaussian distribution of new states and corresponding weights.
(5) Calculating mean and variance of new Gaussian distribution.

3.1.4 Cubature Kalman Filter (CKF): 2009

CKF is a nonlinear filter for high-dimensional state estimation. CKF is a spherical-radial cubature rule which makes it possible to numerically compute multivariate moment integral encountered in the nonlinear Bayesian filter [13]. A third-degree spherical-radial cubature rule that provides a set of cubature points scaling linearly with the state-vector dimension is driven. The CKF is better than EKF and UKF in convergence and dimensions and it is useful for calculating the cost because it does not have a derivation [14].

CKF has high stability and accuracy. CKF is used for approximating a n-dimensional Gaussian integral. CKF selects $n2$ cubature points based on spherical-radial rule, n

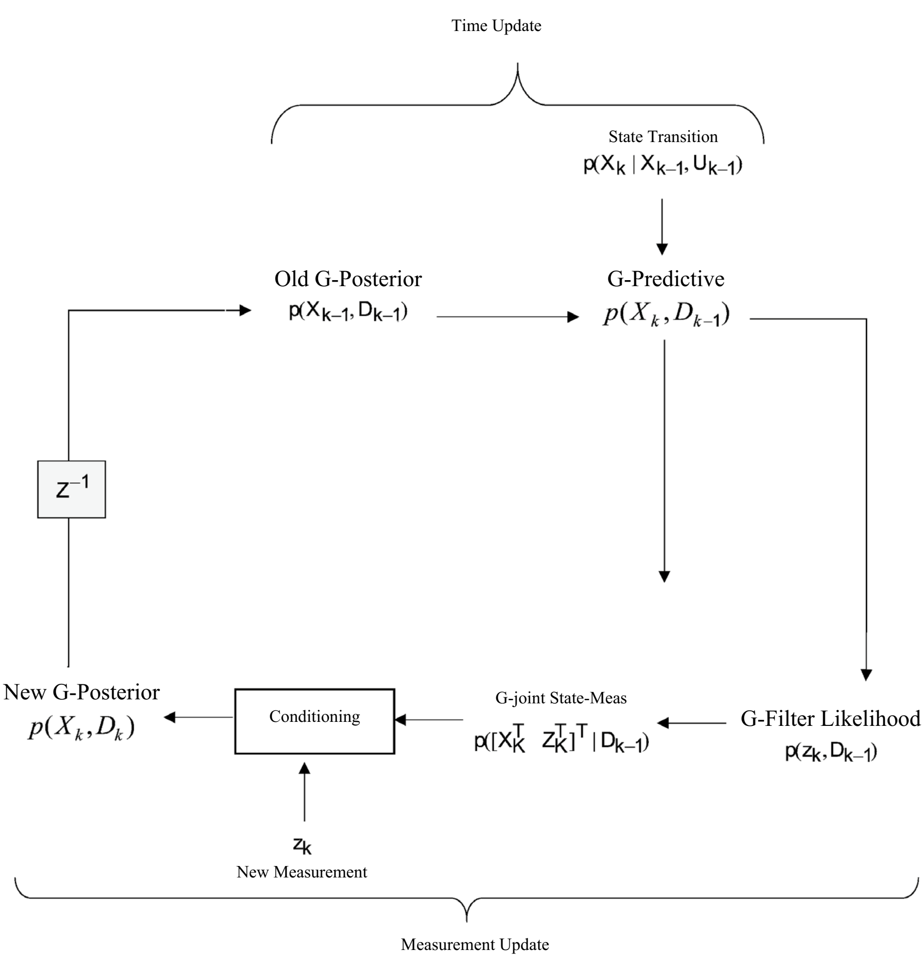


Fig. 3 Signal-flow diagram of the recursive Bayesian filter under the Gaussian assumption, where "G-" stands for "Gaussian-" [13]

Table 1 Kalman filtering framework [13]

$P_{zz,k\|k-1}$	Innovation covariance
$(z_k - \hat{z}_{k\|k-1})$	Innovation
W_k	Kalman gain

shows state dimensions, in which i is the second point that is located in spherical intersection and its axes, it is related weight to cube point. Therefore, it can be understood that CKF uses $n2$ scores, that is probably calculated by Gauss integral [15] (Fig. 3).

The key approximation taken to develop the Bayesian filter theory under the Gaussian domain is that the predictive density $p(X_k|D_{k-1})$ and the filter likelihood density $p(Z_k \; D_k)$ are both Gaussian, which eventually leads to a Gaussian posterior density $p(X_k|D_k)$.

Under the Gaussian approximation, the functional recursion of the Bayesian filter reduces to an algebraic recursion operating only on means and covariance's of various conditional densities encountered in the time and the measurement updates:

Time Update

In the time update, the Bayesian filter computes the mean $\hat{X}_{k \; k-1}$ and the associated covariance $P_{k|k-1}$ of the Gaussian predictive density as follows:

$$\hat{X}_{k \; k-1} = E[X_k \; D_{k-1}], \tag{7}$$

where E is the statistical expectation operator.

$$\begin{aligned}\hat{X}_{k|k-1} &= E[f(X_{k-1},\ U_{k-1})D_{k-1}] \\ &= \int_{R^{n_x}} f(X_{k-1},\ U_{k-1})p(X_{k-1},\ U_{k-1})dX_{k-1} \\ &= \int_{R^{n_x}} f(X_{k-1},\ U_{k-1}),\end{aligned} \tag{8}$$

$$\times N(X_{k-1}; \hat{X}_{k-1 \;\; k-1}, P_{k-1 \;\; k-1})dX_{k-1} \tag{9}$$

where $N(.,.)$ is the conventional symbol for a Gaussian density.

Measurement Update

On the receipt of a new measurement z_k, the Bayesian filter computes the posterior density $p(X_k|D_k)$ yielding.

$$p(X_k|D_k) = N(X_k; \hat{X}_{k|k}, P_{k \;\; k}) \tag{10}$$

where

$$\hat{X}_{k \;\; k} = \hat{X}_{k|k-1} + W_k(z_k - \hat{z}_{k|k-1}) \tag{11}$$

$$P_{k \;\; k} = P_{k|k-1} - W_k P_{z \; z,k \;\; k-1} W_k^T \tag{12}$$

$$W_k = P_{x \; z,k \;\; k-1} P^{-1}_{z \; z,k \;\; k-1^0} \tag{13}$$

If f(0) and h(0) are linear functions of the state, the Bayesian filter under the Gaussian assumption reduces to the KF.

Table 1 shows how quantities derived above are called in the Kalman filtering framework. The signal-flow diagram in Fig. 4 summarizes the steps involved in the recursion cycle of the Bayesian filter [13].

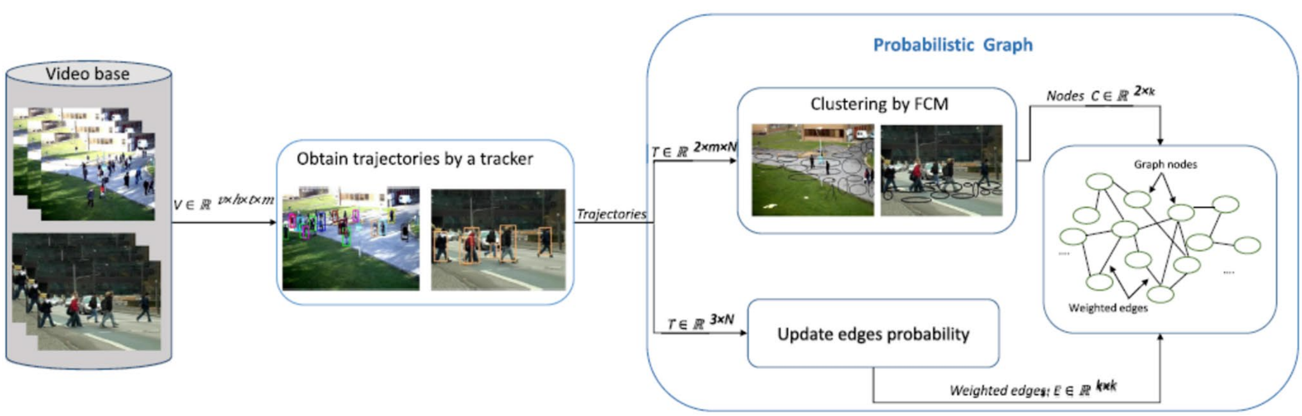


Fig. 4 This figure presents the process of building the probabilistic graph. First, targets in videos are tracked by an obituary tracker and then classified by FCM technique. The obtained centers by FCM are the graph nodes. Then, for each successive points of a trajectory, the weight between two similar nodes is increased by one [16]

3.1.5 Probabilistic Kalman Filter (PKF): 2020

PKF was presented in 2020 that is able to take into account the stored trajectories from the same environment. It has been successfully applied to tracking moving objects in real time situations. PKF is able to take into account the stored trajectories to improve tracking estimation. The PKF has an extra stage after two steps of the KF to refine the estimated position of the targets [16]. Here, A new model is created by applying the fuzzy C-means to the stored trajectories according to which a probabilistic graph is constructed, summarizing the overall moving of objects in the environment. The smoothing step of the KF can then be activated by finding the most probable future path of a moving object applying the Viterbi algorithm [17, 18] to the constructed graph. The graph is built in the offline situation and could be adapted in the online tracking. The proposed tracker has higher accuracy compared to the standard KF and could handle widespread problems such as occlusion. Another significant achievement of the proposed tracker is to track an object with anomalous behaviors by drawing an inference based on the constructed probabilistic graph [16]..

3.1.5.1 Probabilistic Graph Construction The number of cluster (nodes) is totally depends on the videos and objects. In order to complete the graph, initially, all the nodes are connected to each other which are weighted zero. Then, for each successive points in each stored trajectory, two nodes of the graph which have the minimum distance (i.e., Euclidean distance) to the consecutive points are selected, and the weight of the edge between the corresponding nodes is increased by one. This procedure must be performed over all the successive points in all trajectories. The heaviest edge depicts a crowded location in the environment [16] (Fig. 5).

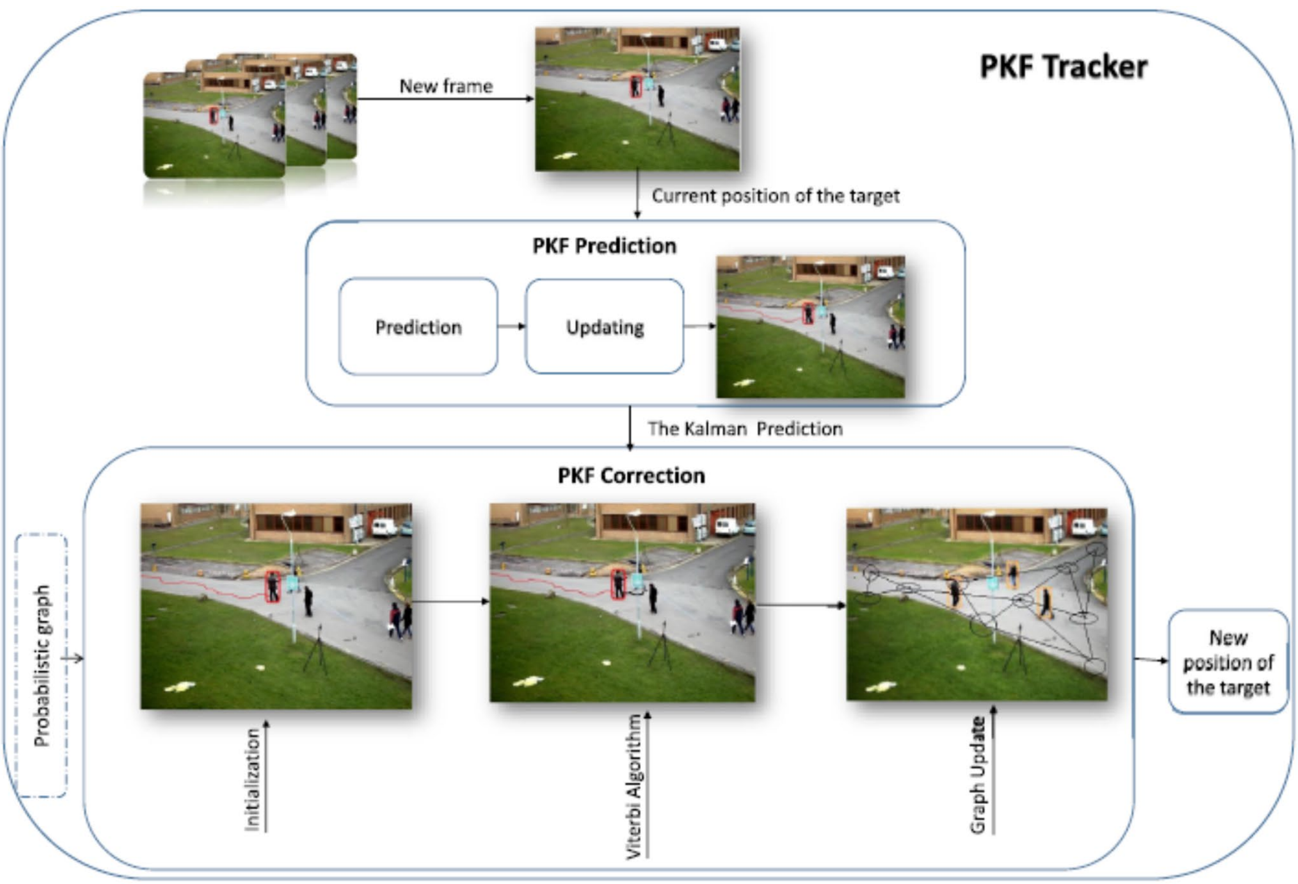


Fig. 5 The flowchart shows the PKF construction in one frame. Firstly, the Kalman filter is applied to predict and update the next position. Next, in the initialization step of the probabilistic Kalman filter, the most similar node of the graph to the Kalman prediction is found (Initialization step). Then the Viterbi algorithm finds the most probable path of the target as future positions of the target (Viterbi step). Eventually, based on the learning rate, the graph is updated (graph updating step). This process is done for each frame of the video. Algorithm 1 summarizes construction of the probabilistic graph [16]

Algorithm 1 building the probability graph

Require: Screened videos with fixed-camera
1: **for** All videos **do**
2: for All targets in the video do
3: Target detection and obtaining its trajectory
4: Trajectory is stored in the trajectory matrix
5: **end for**
6: **end for**
7: Cluster trajectories by FCM the number of class is C=50 (user-defined)
8: **for** each pair points in a trajectory matrix **do**
9: the Weight between two similar nodes is increased by one
10: **end for**
11: **return** A probabilistic graph

3.1.5.2 Graph Updating Algorithm 2 shows the overall steps for the graph updating.

Algorithm 2 The graph updating phase

Require: Setting the learning term λ, y_{obs}, the probabilistic graph
1: Updating winning node Vi of the path
2: Computing $v_{i,k} = v_{i,k-1} + (1 + \frac{1}{1+\lambda})\ (yobs - vi{,}k{-}1)$
3: **return** Probabilistic graph

3.1.5.3 Time Complexity Algorithm 3 shows the entire process of the PKF in the real-time system.

Algorithm 3 The PKF

Require: Detective of the target in the first frame.
1: Initialize: Set the Kalman filter parameters (H and F).
2: **for** i = 2 to the number of frames **do**
3: The prediction and correction by the Kalman filter.
4: The PKF parameters initialization.
5: Find the most probable path of the graph by the Viterbi: $S^* = \arg\max_s P(S \mid m_i^s)$
6: **for** $j = 1$ *to length of* S *do*
7: $\lambda_k = \sum_k H_k^T (H_k P_k H_k^T + R_k)$
8: $\lambda_{k,k+1} = \hat{x}_{k,k+1} + \lambda(S[j] - H_k \hat{x}_k)$
9: $\sum_{k+1} = \sum_k (F_k - L_k H_k)^T$
10: **end for**
11: **end for**
12: Graph updating based on Algorithm 2.
13: **Return** The next position of the target

---

4 Types of Multiple Model Kalman Filter

Issues such as parameter uncertainty, not being convergent and sometimes high computational costs can make the model encounter new challenges. Researchers have designed different models for solving problems and current challenges. In the present study, recent studies on MM filters have been studied and categorized. As it can be seen in Fig. 1 and figures below, KFs have been used in problems solving in three general groups, including Basic KFs (Fig. 1), MM Attitude Estimation (MMAE) (Fig. 6) and Interacting MM KF based algorithm (Fig. 7).

Some presented methods have been shown in these groups.

In the following, MMs that have been designed based on basic models, will be introduced.

4.1 Multi-model Filters

When system model and noise are known, KF and EKF are usually performed for obtaining optimal estimation of the state. But in some practical applications, system models cannot be obtained accurately. Providing uncertainty about model, filter performance may decrease [8]. In complicated systems in which behavior modeling is not easy, several filters (each with a different model of system) can work in parallel, after that for obtaining more reliable estimations, outputs of each filter work [19].

We aim at reviewing MMAE [20] and IMM [21]. MMAE algorithm is an effective method for dealing with problems of model uncertainty. In MM, uncertainty is approximated by a set of models. As it is shown in Fig. 8, MMAE uses a parallel bank of KF to present several estimations in which each filter related to a model in a set of models has been predetermined. Final state estimation and covariance is presented by weighted sum of each filter estimation. During past years, MMAE algorithms have been implemented successfully in different programs such as target tracking

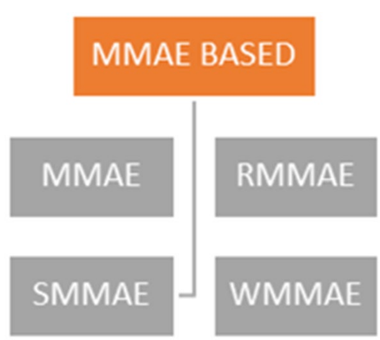


Fig. 6 Category for multiple model attitude estimation (MMAE)

Fig. 7 Category for interacting multiple model KF based algorithm (IMM)

[22], error detection [23], and Calibration bias [24]. Three generations of MMAE have been determined. The first presented algorithm is KF [25]. The second one is IMM [26] and the third algorithm is Variable-Structure MM (VSMM) [27]. In classic algorithm is supposed model is not different from time. Jumps in system model are paid attention to in IMM algorithm. In VSMM, model set is supposed variable and becomes adaptable based on measurements. In MMAE uncertainty in model parameter is supposed. For implementing MMAE, a set of discrete models is created by selecting some points in limited zone. Accordingly, approximation is presented. The most probable parameter in the set of discrete model may not be equal to real model parameter. When the number models are more or denser to cover limited zone by discrete points, the approximation will be more accurate. Many models are needed to achieve a good discretization level. Accordingly, utilizing more models increase computational burden significantly. When model set is big, MMAE will be inapplicable fiscally. However, MMAE has been designed against uncertainty of model, when an unstable model set is used for calculating uncertain model parameter, some errors of parameter detection will be remained. The amount of error in detecting parameter is related to density of discrete elements in model set [19].

4.1.1 Multiple Model Adaptive Estimation (MMAE)

A bank of KFs was first used by [25], as a parallel set of estimators for a sample stochastic process [20]. Improved the technique, calling it MMAE [25, 28–32] and using it to reliably detect and identify sensor and actuator failures in aircraft. MMAE consists of a bank of parallel KFs, each with a different model, and a hypothesis testing algorithm [33]. This modified algorithm, denoted the residual correlation KF bank (RCKFB), uses the magnitude of an estimate of the correlation of the residual with a slightly modified version of the usual MMAE hypothesis testing algorithm to

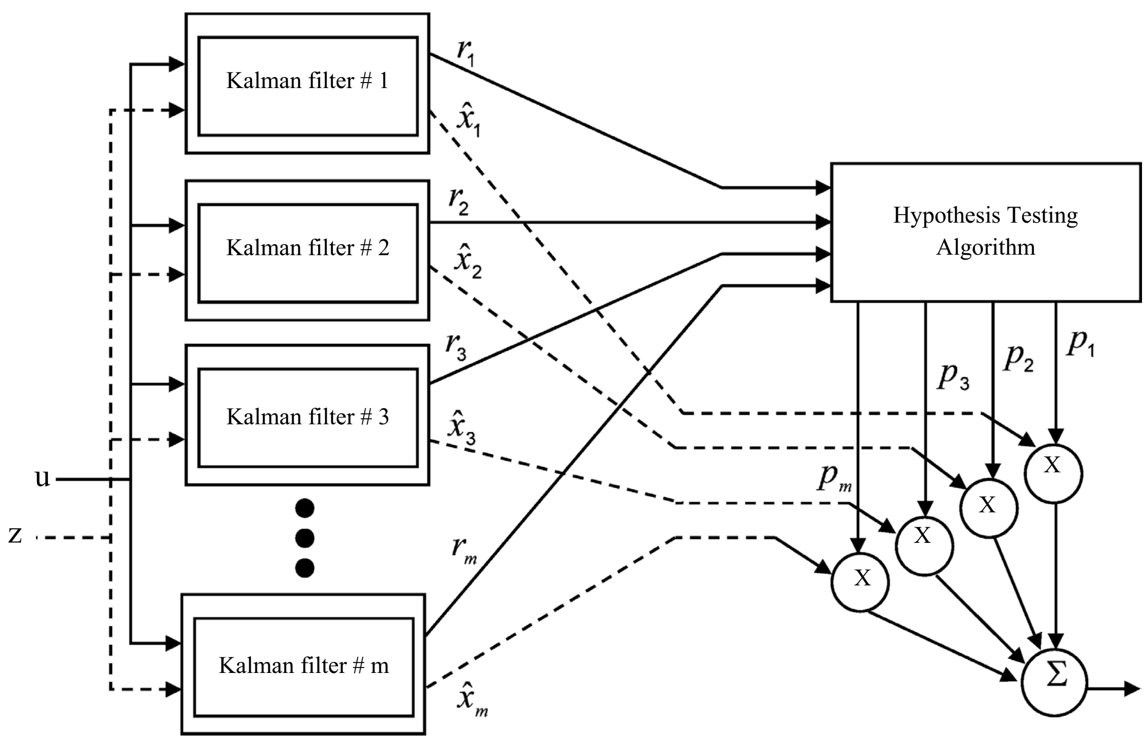


Fig. 8 MMAE Algorithm structure

assign the conditional probabilities to the various hypotheses that are modeled in the KF bank within the MMAE [20]. This concept is used to detect flight control actuator failures, where the existence of a single frequency sinusoid (which is highly time correlated) in the residual of an elemental filter within an MMAE is indicative of that filter having the wrong actuator failure status hypothesis [20]. This technique results in a delay in detecting the flight control actuator failure because several samples of the residual must be collected before the residual correlation can be estimated. However, it allows a significant reduction of the amplitude of the required system inputs for exciting the various system modes to enhance identifiably, to the point where they may possibly be subliminal, so as not to be objectionable to the pilot and passengers.

MMAE [25, 28–32] consists of a bank of parallel KFs, each with a different model [20]. One of MMAE applications is flight control sensor/actuator failure detection and identification which each KF has a different failure status model (a_k) [20]. The MMAE compares the magnitudes of the residuals (appropriately scaled to account for various filters and chooses the hypothesis that corresponds to the residual that has a history of having smallest (scaled) magnitude [20]. Large residual must be produced by the filters with models that are incorrect to be able to discount these incorrect hypotheses. The residual is the difference between the measurement of the system output and the filter's prediction of what that measurement should be, based on the filter-assumed system model [20]. Residual spectral content estimation is used by hypothesis testing algorithm in which corrected hypothesis testing algorithm can be interpreted as residual filter by Band-pass filter with f as center, 3 decibel of band width, $N=1$, output sampling and calculating square root [34]. N = 1 coefficient is needed for estimating needed power spectral density. It is obvious that by more data (N increases) band width becomes narrow and much noise passes out of band width. For cases in which residual has non-zero mean, this estimation works well. Generally, when motion failure happens, input elements of control are filtered by incorrect hypothesis. If the control input elements are sinusoids, then the residual includes a sinusoid, with the same frequency as the control input elements. Therefore, content of the residual can be used for the spectral content of the residual, at the input frequencies, to indicated the presence of an actuator failure. When the control input is fairly strong, this technique takes longer than the standard MMAE to make the correct failure identification, due to the time used to collect a sufficient number of samples to generate an adequate spectral estimate. However, this technique provides failure identification performance at small input levels where the standard MMAE no longer functions [20].

4.1.2 RMMAE Algorithm

Reference [19] have introduced Robust Multiple Model Adaptive Estimation. RMMAE has been formulated for nonlinear systems with parameter uncertainty that is a mixture of MMAE [20] and RKF [35].

Each cycle of RMMAE algorithm exists in three stages:

Stage 1 Parallel filtering in time k, each filter predicts and updates its status and covariance separately, it is assumed that parameter vector θ (τ) corresponds parameter vector real θ.

Stage 2 Weight update, obtained status estimation weight from separate filters is calculated by residual $\tilde{\mathbf{y}}_k^{(\tau)}$ and related covariance $\tilde{\mathbf{\Omega}}_k^{(\tau)}$

$$\omega_k^{(\tau)} = \frac{\omega_{k-1}^{(\tau)}\Lambda_k^{(\tau)}}{\sum_{\tau=1}^{M}\omega_{k-1}^{(\tau)}\Lambda_k^{(\tau)}} \tag{14}$$

$$\Lambda_k^{(\tau)} = \frac{1}{\sqrt{2\pi\hat{\Omega}_k^{(\tau)}}}\exp\left[-\frac{1}{2}\overline{y}_k^{(\tau)T}(\hat{\Omega}_k^{(\tau)})^{-1}\overline{y}_k^{(\tau)}\right] \tag{15}$$

$$\overline{y}_k^{(\tau)} = y_k - h(\hat{x}_{k|k-1}^{(\tau)}, \theta^{(\tau)}) \tag{16}$$

$$\hat{\Omega}_k^{(\tau)} = H_k(\theta^{(\tau)})P_{k|k-1}^{(\tau)}H_K^T(\theta^{(\tau)}) + \hat{R}_k \tag{17}$$

Stage 3 Combination of states estimation of RMMAE is parallel to weighted sum of filter estimation $\tilde{\chi}_k^{(\tau)}$.

$$\hat{x}_k = \sum_{\tau=1}^{M}\omega_k^{(\tau)}\hat{x}_k^{(\tau)} \tag{18}$$

Covariance of state estimation error is calculated as follow:

$$P_k = \sum_{\tau=1}^{M}\omega_k^{(\tau)}[P_k^{(\tau)} + (\hat{x}_k - \hat{x}_k^{(\tau)})(\hat{x}_k - \hat{x}_k^{(\tau)})^T] \tag{19}$$

This algorithm is a static algorithm with multiple- model in which it is assumed that model is not different from time. It is considered for XNAV system. The difference between RMMAE and MMAE is that RKF is used for implementing parallel filtering. RKF is used as a sub-filter. It is claimed that RKF guarantees an upper limit for covariance of estimation error, computational cost of RKF is almost similar to EKF. The main advantage of RMMAE is its resistance to error detection of model parameter. Suggested algorithm for XNAV is performed in presence of error of pulsar angular position. The result of simulation shows that RMMAE is better than EKF, UKF, MMAE, and RKF in terms of spacecraft position. Furthermore, RMMAE performance is more stable than RKF, when uncertain parameters are formulated as different values.

4.1.3 SMMAE Algorithm

Suggested scheme by [36] combines Innovation-based Adaptive Estimation (IAE) [37] and MMAE [38] in a structure, state estimation for each model is performed by IAE. Attitude estimation of a vehicle that is roll(Φ), pitch(Φ), and angle (ψ) has been performed in presence of inertia sensor from the perspective of Attitude and Heading Reference system. This system consists of inertia 3-axial sensors such as accelerometer, Gyroscope, and Magnetometer that are used for measuring linear acceleration, angular velocity, and heading for a moving vehicle respectively. The key idea in SMMAE is to start with a large number of models by allocating equal weight to each model and deleting those with low weighting coefficient. Due to these models accuracy increases and finally the most effective models are selected. The results show an improvement in performance than MMAE. Generally, pseudo code for implementing SMMAE is as follow:

(1) Consider 50 different filters in filter stack of SMMAE.
(2) Initialization of system states and noise covariance matrixes Q & R is distributed around a fix value.
(3) First, put the weight 1/5 m for each filter.
(4) Calculate EKF estimation in stack and weighting coefficient of innovation.
(5) Estimate system state as weighting estimation of separate filters.
(6) Make it unload whenever its weight reaches under 0.0001.
(7) Whenever the number of current filters reaches 5, stop unloading.
(8) Repeat normal MMAE with 5 final filter in stack (Fig. 9).

4.1.4 WMMAE Algorithm

WMMAE is another suggested method by [39], WMMAE is a window based MMAE. The main goal of this method is to improve state estimation by incorporating window size as one of the unknown parameters in MMAE framework, referred to as window based MMAE (WMMAE) [39]. The proposed scheme chooses the best filter from filter bank using MMAE feature in which each filter has a different size of window. For example, WMMAE scheme an extra unknown parameter introduces window size in MMAE framework. Used formulas for updating R and Q matrixes in each filter block are the same. Obtained estimation error of each current filter in filter bank is used for calculating weighting coefficient for individual IAE blocks in WMMAE structure. IAE and MMAE are mixed in a structure and state estimation is performed for each model by IAE. IAE is one of adaptive KF designs that uses KF in its structure, it is preferred because of its simplicity. In this method, IAE is used in MMAE instead of EKF. In

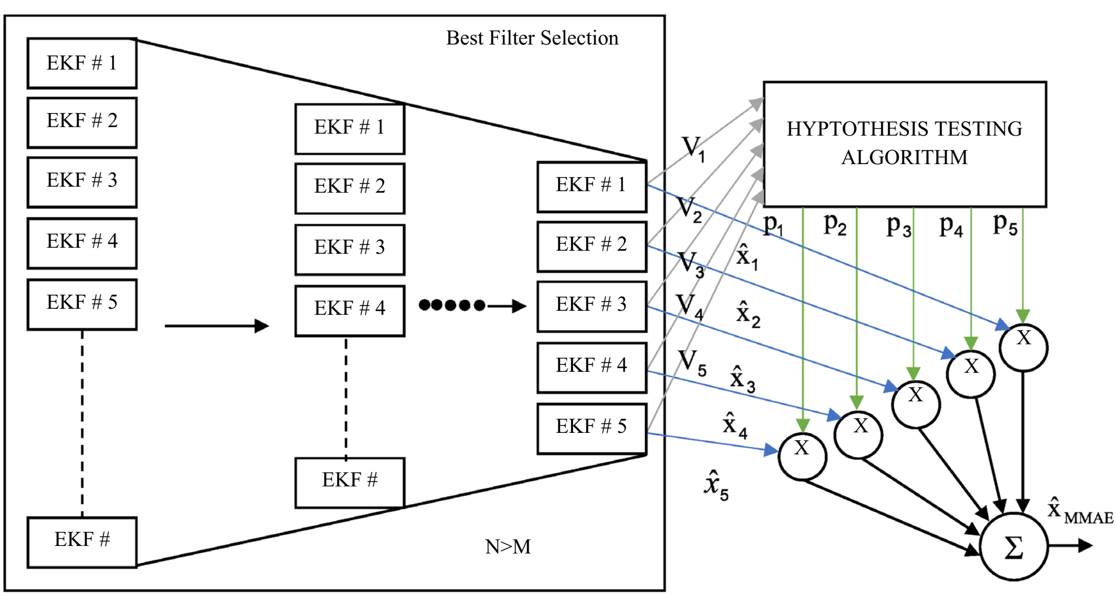


Fig. 9 SMMAE structure

fact, IAE uses EKF in an internal framework for adapting process and measurement of noise covariance parameter (Q and R) based on innovation or residual sequence, or innovation resulted from data. In order to estimate attitude using inertia sensors, measurement process and model are used for state estimation in KF framework. For obtaining attitude and title of a vehicle by state vector of EKF as enforcing 3 Euler angles (ψ, ‘ θ‘ ϕ) by Gyroscope with fixed value (b1r, b1q, b1p) measurement process and model of an EKF have been formulated. Process model for AHRS system has been explained using the following equations:

Total model can be summarized as follow:

$$\dot{\Psi}(\tau) = q(\tau)\frac{s\phi}{c\theta} + r(\tau)\frac{c\phi}{c\theta} + b_{1q}(\tau)\frac{s\phi}{c\theta} + b_{1r}(\tau)\frac{c\phi}{c\theta} + w_q\frac{s\phi}{c\theta} + w_r\frac{c\phi}{c\theta}, \tag{20}$$

$$\dot{\theta}(\tau) = q(\tau)c_\phi - r(\tau)s_\phi + b_{1q}(\tau)c_\phi - b_{1r}(\tau)s_\phi + w_q c\varphi - w_r s\phi, \tag{21}$$

$$\begin{aligned}\Phi(\tau) &= p(\tau) + q(\tau)\frac{s\phi s\theta}{c\theta} + r(\tau)\frac{c\phi s\theta}{c\theta} + b_{1p}(\tau) + b_{1q}(\tau)\frac{s\phi s\theta}{c\theta}\\ &+ b_{1r}(\tau)\frac{c\phi s\theta}{c\theta} + w_p + w_q\frac{s\phi s\theta}{c\theta} + w_r\frac{c\phi s\theta}{c\theta},\end{aligned} \tag{22}$$

$$\begin{aligned}\dot{b}_{1p} &= w_{1p}\\ \dot{b}_{1q} &= w_{1q}\\ \dot{b}_{1r} &= w_{1r}.\end{aligned} \tag{23}$$

Here p(t), q(t) and r(t) are the angular rates measured by gyroscope along x, y and z axes respectively, S_ψ, θ, φ, C_ψ, θ, φ and T_ψ, θ, φ represent the sine, cosine and tangent of the specific angle in the subscript, respectively and W_p, W_q, W_r, $W1_p$, $W1_q$ and $W1_r$ are approximated as zero mean Additive White Gaussian Noise (AWGN).

Proposed scheme of WMMAE is applied on attitude estimation that system model has been defined for it above. State estimation framework of IAE deals with the problem, it gives much better result compared to other estimation schemes, it uses both forms of adaptive KF and improves state estimation greatly, whereas this scheme is not dependent to any feeding parameters, it can be optimal based on data converting to filter. Obtained state estimation from each stacked filter of WMMAE is combined in a probable framework for producing an optimal output. One of the most important contributions of this scheme is including IAE in filter stack instead of normal EKF, resulting in robust system state output in each period. Experimental results have confirmed the effect of proposed scheme on creating an adaptive system resistance to system noise with different window size. WMMAE performance is tested by attitude measurement, one of the key subjects in IAE is to select appropriate window size for a certain route. Although some corrections have been done on IAE, they are only limited to deal with outliers. Total pseudo code of WMMAE implementation is as follow:

WMMAE algorithm

(1) Consider different filters in packed filter of WMMAE.
(2) Initialize system states and noise covariance matrixes of Q and R.
(3) Implement normal IAE for several times in order to obtain convergent values for Q and R.
(4) Regulate these convergent values of Q and R for each filter in stacked filter of WMMAE with different window sizes.
(5) Provide a balance factor q/m for each filter in stack of WMMAE.
(6) Estimating system concerning MMAE equations in article with different window sizes for IAE in stack filter.
(7) Calculate the amount of innovation for each filter in stack.
(8) Updating noise covariance matrix based on IAE equations and relative weight of filters by MMAE equations.
(9) Estimating final state of system as weighting sum from each filter output.
(10) Go to next stage, then go back to stage 6 (Fig. 10).

4.2 Kalman Filter Based on IMM

In IMM by decreasing complexity via combining initial conditions of each filter in each stage and obtaining delays of lower state estimation, MMAE is improved [40]. IMM has been designed especially for exact tracking of targets that their state and measurement change while moving. When these models are nonlinear, IMM algorithm should be corrected to guarantee the exact route. IMM is one of rare algorithms that is able to overcome problems of model changes while moving, it can change from one model to another [14].

Figure 11 shows the IMM structure [41, 42]

4.2.1 IMMCKF Model: 2010

Reference [14] suggested CKF should be used in IMM algorithm instead of UKF. In this method, IMM algorithm CKF based (IMM-CKF) was presented to tracking maneuvering target with angle-only measurements.

CKF is about finding a set of points and weight for numerical analysis of two initial states (Mean and Covariance). CKF uses cubic spherical-radial function for finding points and weights. Both steps including update and measurement are calculated at any stage of time. This algorithm stages are as follow (Fig. 12):

Interaction

Starting with $\hat{x}^{0j}_{k-1|k-1}$ and $P^{0j}_{k-1|k-1}$, we compute the mixed initial condition for the filter matched to $\mathbf{M}^{(}_{k}j)$:

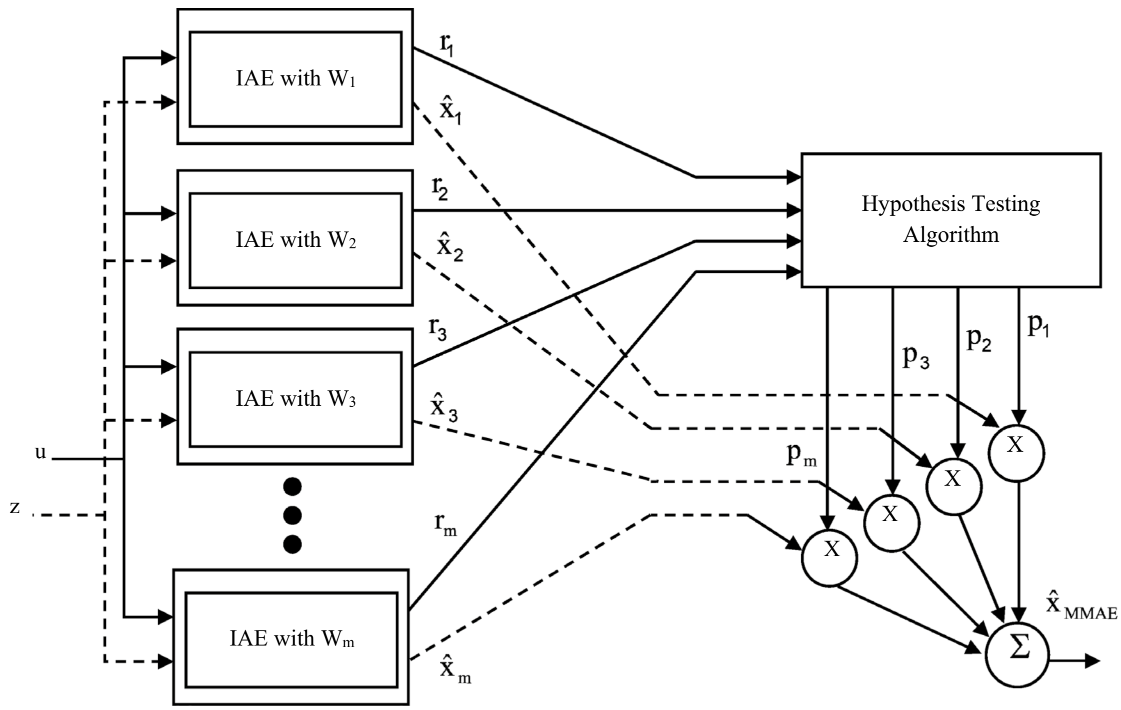


Fig. 10 WMMAE structure

$$\hat{x}^{0j}_{k-1|k-1} = \sum_{i-1}^{r} \hat{x}^{i}_{k-1|k-1}\mu^{i|j}_{k-1|k-1} \qquad i,j = 1, ..., r \tag{24}$$

$$P^{0j}_{k-1|k-1} = \sum_{i=1}^{r} \mu^{i|j}_{k-1|k-1}, \left\{P^{i}_{k-1|k-1} + \left[\hat{x}^{i}_{k-1|k-1} - x^{0j}_{k-1|k-1}\right], \left[\hat{x}^{i}_{k-1|k-1} - x^{0j}_{k-1|k-1}\right]^{T}\right\} \quad i,j = 1, \dots, r \tag{25}$$

$$\mu^{i|j}_{k-1|k-1} = \frac{1}{c_j} p_{ij}\mu^{i}_{k-1} \qquad i,j = 1, ..., r \tag{26}$$

is the probability that model $\mathbf{M}^{(}I)$ was in effect at k − 1 given that $\mathbf{M}^{(}j)$is in effect at time k conditioned on z_{k-1}. And where $\mathbf{M}^{(}_{k-1}I)$is the probability of mode $\mathbf{M}^{(}I)$at time k-1.

(1) Filtering based on the model.
Above estimation and covariance are used as CKF input in accordance with M_j.
(2) Probable updating of the model

$$\Lambda^{j}_{k} = \frac{1}{\sqrt{\left|2\pi S^{j}_{k}\right|}} \exp\left[-\frac{1}{2}(v^{j}_{k})' S^{j}_{k} v^{j}_{k}\right] \tag{27}$$

Model probabilities of innovation of filter are updated.

Probability functions related to filters r are calculated by:

(3) Combination of estimation and estimation covariance and output covariance are calculated concerning combined equations:

$$\hat{x}_{k|k} = \sum_{j=0}^{r} \mu^{j}_{k}\left\{P^{j}_{k|k} + \left[\hat{x}^{j}_{k|k} - \hat{x}_{k|k}\right]\left[\hat{x}^{j}_{k|k} - \hat{x}_{k|k}\right]^{T}\right\} \tag{28}$$

In order to test IMM-CKF performance, IMM-EKF and IMM-UKF are used. The results show that IMM-CKF has a better performance in estimation of location and velocity. Three IMM based filters such IMM-EKF, IMM-UKF, and IMM-CKF are compared concerning performance and computational complexity. Simulation results show that IMM-CKF has the best performance in decreasing estimation errors, whereas IMM-EKF as a working method for tracking maneuvering target has the worst performance. Whenever estimation errors are important, IMM-CKF is an appropriate option for a better performance. If computational laid is important, IMM-EKF should be chosen for computational performance.

4.2.2 EKF-SIMM-EKF-MIMM-UKF-SIMM-UKF-MIMM: 2012

Reference [43] suggested that by using EKF and UKF, these filters improve for nonlinear systems.

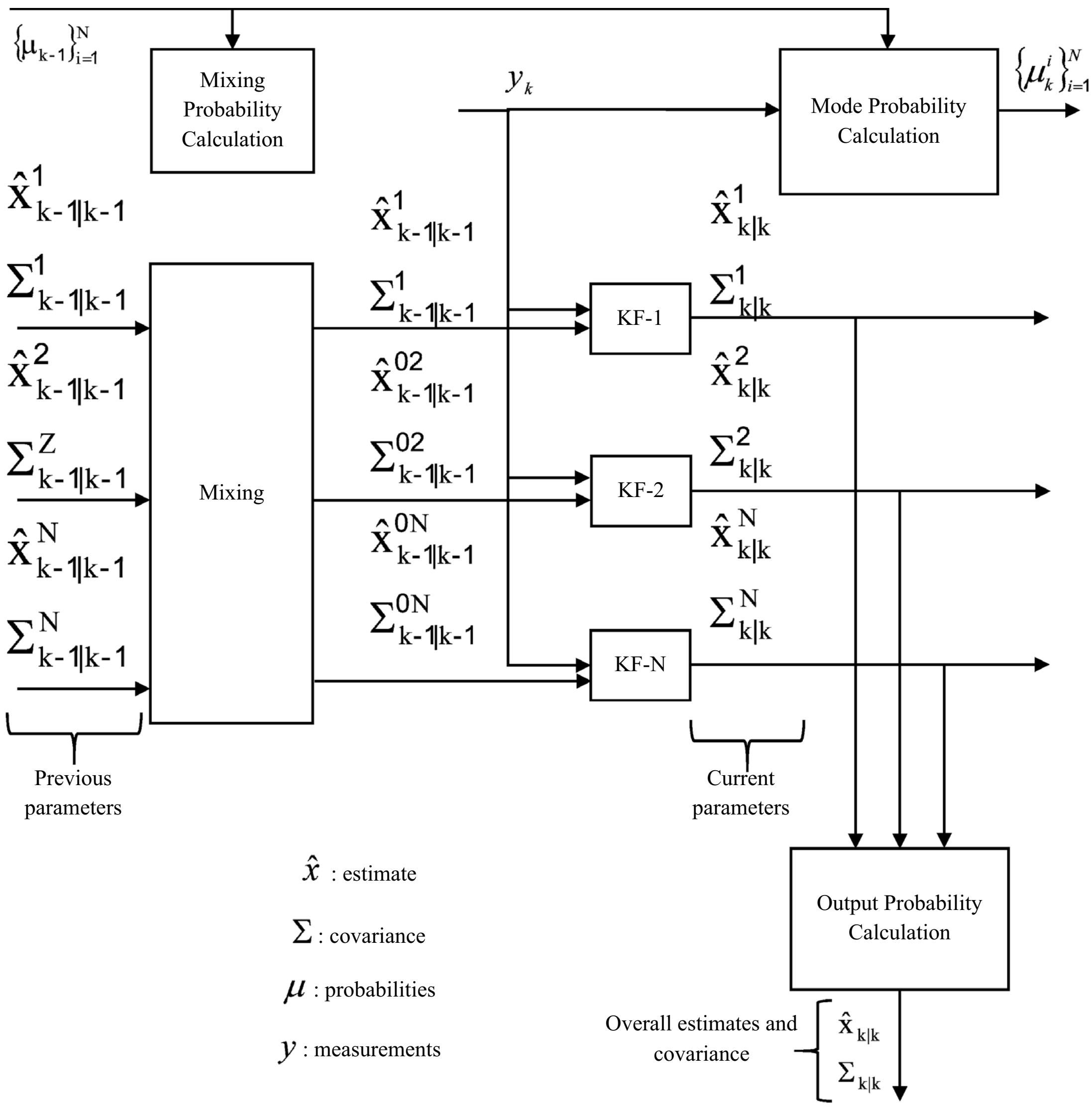


Fig. 11 IMM process

EKF-SIMM-EKF-MIMM-UKF-SIMM-UKF-MIMM are introduced as improved algorithms of IMM and their performance is compared. It is claimed that proposed improved algorithms can receive estimation of target optimal state in the least nonlinear variance. By comparing EKF-IMM and UKFIMM, the effectiveness of proposed algorithms is confirmed. Proposed algorithms are the best in speed estimation.

4.2.2.1 **UKF-SIMM Algorithm**

Step 1 Calculating combined scale weight for filter in accordance to model.

$M_j^k (j \in S = \{1, 2, \ldots s\})$

$$P\left\{M_i^{k-1}|M_j^k, Z^{k-1}\right\} = \frac{\pi_{ij} a_i^{k-1}}{\bar{c}_j} \qquad i,j \in S \tag{29}$$

When S is a set of models, and s is the number of elements of the set, M_{jk} represents model j in k time.

$$\pi_{ij} = prob\left\{M_j^k|M_j^{k-1}\right\} = P\left\{M_j^k|M_j^{k-1}\right\},$$
$$a_i^{k-1} = prob\left\{M_j^k|Z^{k-1}\right\} = P\left\{M_i^{k-1}|Z^{k-1}\right\},\ \bar{c}_j = \sum_{i=1}^{m} \pi_{ij}\mu_i(k-1) \tag{30}$$

Step 2 Calculating initializing state X_{oj} and related covariance P_{oj} for filter.

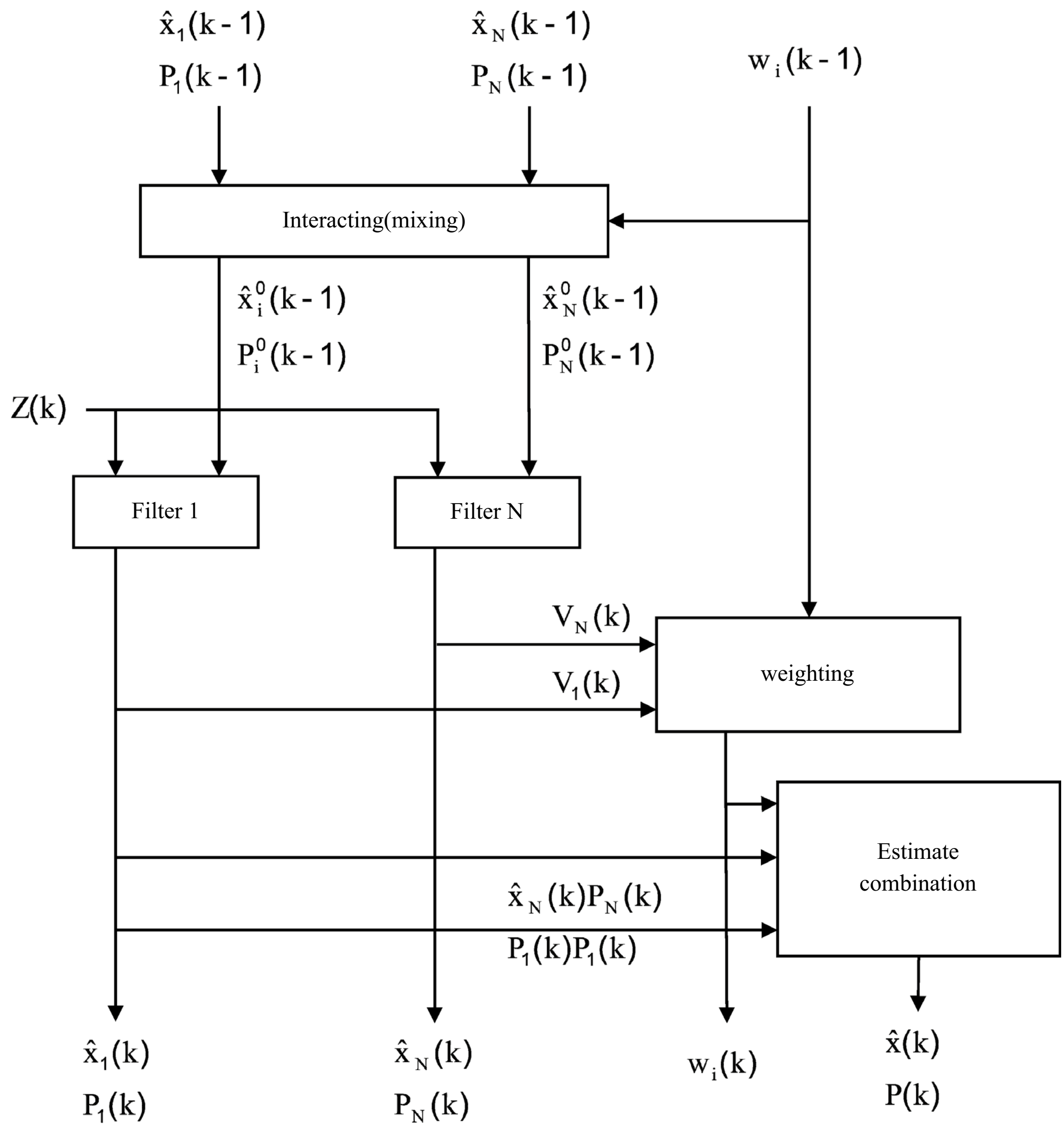


Fig. 12 IMM-CKF structure

$$\hat{x}_{0j}(k|k) = \sum_{i=1}^{s} a_{i|j}(k|k)\hat{x}_{j\_s\_ukf}^{k-1} \tag{31}$$

$$P_{0j}(k|k) = \sum_{i=1}^{s} a_{t|j}(k|k)\{P_{j\_s\_ukf}^{k-1} + [\hat{x}_{j\_s\_ukf}^{k-1} - \hat{x}_{0j}(k|k)] \times [\hat{x}_{j\_s\_ukf}^{k-1} - \hat{x}_{0j}(k|k)]^T\} \tag{32}$$

Step 3 Filtering UKF using algorithm produces output.

Step 4 Combining state estimation and related covariance concerning updated scale weight.

$$\hat{x}_{s\_ukf}(k) = \sum_{i=1}^{s} a_j^k \hat{x}_{j\_s\_ukf}^k \tag{33}$$

$$a_j^k = \left(\sum_{i=1}^{s} \frac{1}{trP_{j\_s\_ukf}^k}\right)^{-1} \frac{1}{trP_{j\_s\_ukf}^k} \tag{34}$$

$$P_{s\_ukf} = \sum_{j=1}^{s} (a_j^k)^2 P_{i\_s\_ukf}^k \tag{35}$$

4.2.2.2 EKF-SIMM Algorithm EKF SIMM algorithm has the same process as UKF-SIMM except in step 3. It means state estimation and related covariance are calculated using outputs of EKF production and EKF-DIMM algorithm. Then, final combination of outputs in step 4 is calculated as follow:

$$\hat{x}_{s\_ekf}(k) = \sum_{j=1}^{s} a_j^k \hat{x}_{j\_s\_ekf}^k,\ a_j^k = \left(\sum_{i=1}^{s} \frac{1}{trP_{i\_s\_ekf}^k}\right)^{-1} \frac{1}{trP_{i\_s\_ekf}^k},\ P_{s\_ekf} = \sum_{j=1}^{s} (a_j^k)^2 P_{i\_s\_ekf}^k \tag{36}$$

4.2.2.3 UKF-MIMM Algorithm

Step 1 Calculating weight of mixed matrix for correspondent filter to Mjk model.

$$\Omega_{i|j}(k|k) = p\{M_i^{k-1}|M_i^k, Z^k\} = \frac{\pi_{ij}\Omega_i^{k-1}}{\sum_{i=1}^{s}\pi_{ij}\Omega_i^{k-1}} = (\sum_{i=1}^{s}\pi_{ij}\Omega_i^{k-1})^{-1}\pi_{ij}\Omega_i^{k-1} \tag{37}$$

Step 2 Calculating initial mixed state of Xoj and related covariance P_{oj} for adapting filter and X_{oj} model.

$$\hat{x}_{0j}(k|k) = \sum_{t=1}^{s} \Omega_{i|j}(k|k)\hat{x}_{j\_m\_ukf}^{k-1} \tag{38}$$

$$P_{0j}(k|k) = \sum_{t=1}^{s} \Omega_{i|j}(k|k)\left\{P_{i\_m\_ukf}^{k-1} + \left[\hat{x}_{j\_m\_ukf}^{k-1}\hat{x}_{0j} - (k|k)\right] \times \left[\hat{x}_{j\_m\_ukf}^{k-1} - \hat{x}_{0j}(k|k)\right]^T\right\} \tag{39}$$

Step 3 Producing algorithm output by UKF filters.

Step 4 Combining state estimation and relayed covariance concerning updated matrix weight.

$$\hat{x}_{m\_ukf}(k) = \sum_{j=1}^{s} \Omega_j^k \hat{x}_{j\_m\_ukf}^k \tag{40}$$

Updated matrix weight of M_{jk} model is:

$$\Omega_j^k = \left(\sum_{i=1}^{s}\left(P_{i\_m\_ukf}^k\right)^{-1}\right)^{-1}\left(P_{j\_m\_ukf}^k\right)^{-1} \tag{41}$$

Estimator error variance matrix is optimal fusion.

$$Pm\_ukf = \left(\left(\sum_{i=1}^{s}\left(P_{i\_m\_ukf}^k\right)^{-1}\right)^{-1}\right) \tag{42}$$

4.2.2.4 EKF-MIMM Algorithm EKF-MIMM algorithm has a similar process as UKF-MIMM except in step 3, that is, state estimation $x^(k)_{j\_m\_ekf}$ and related covariance $p^(k)_{j\_m\_ekf}$ are calculated by output of EKF production using EKF-DIMM algorithm. Then, final combination of outputs in step 4 is as follow:

$$\hat{x}_{m-ekf}(k) = \sum_{j=1}^{s} \Omega_j^k \hat{x}_{j\_m\_ekf}^k,\ \Omega_j^k = \left(\sum_{i=1}^{s}\left(P_{i\_m\_ekf}^k\right)^{-1}\right)^{-1}\left(P_{j\_m\_ekf}^k\right)^{-1},\ P_{m\_ekf}^k = \left(\left(\sum_{i=1}^{s}\left(P_{i\_m\_ekf}^k\right)^{-1}\right)^{-1}\right) \tag{43}$$

UKF-MIMM is more accurate than EKF-IMM and UKF-IMM, but EKF-SIMM is better in time. Therefore, proposed algorithm can be competitive options for classic filter algorithm IMM based for nonlinear tracking maneuvering target.

The results of stimulation show that EKF-SIMM, EKF-MIMM, UKF-MIMM algorithms act better in speed estimation compared to EKF-IMM and UKF-IMM. Furthermore, UKF-MIMM has the best estimation accuracy and computational complexity when performance time of EKF-SIMM is the least.

4.2.3 OMTM-IMM Model: 2014

Reference [44] suggest Optimal Mode Transition Matrix IMM to increase accuracy of initial state and state transition.

There are some problems related to IMM algorithm when they are used in tracking target systems. For instance, state transition matrix is incorrect and when time has not been specified it cannot be determined. For solving this problem, an Optimal Mode Transition Matrix (OMTM-IMM) has been suggested that uses the least linear variance theory for calculating Mode Transition Matrix, it depends on continuous system condition not the time. Moreover, alternative filter correlation has been considered. Then, covariance matrixes are used to calculate Mode Transition Matrix. In this algorithm, model probability has been defined as a diagonal matrix that is combined with some filters outputs. Therefore, produced effects by each state can be detected [44].

OMTM-IMM algorithm steps are as follow:

Step 1 Calculate probabilities of optimal mode transition.

$$\pi_{ij}^{opt}(k-1) = \prod_{b\neq j}^{n} \mu_b(k-1) \sum_{a=1}^{n} \alpha_{ja}(k-1) \tag{44}$$

$$\times \left[\sum_{d=1}^{n}\left(\prod_{b\neq d}^{n} \mu_b(k-1) \sum_{a=1}^{n} \alpha_{da}(k-1)\right)\right]^{-1} \quad (i,j,a,b,d \in \mathrm{N}) \tag{45}$$

Step 2 Calculate initial mixed probabilities for j model (I, j…N).

$$\mu_{i|j}(k-1) = \pi_{ij}(k-1)\mu_i(k-1)\overline{c}_j^{-1}. \tag{46}$$

Step 3 Calculate initial mixed probabilities for j model (I, j…N).

$$\begin{aligned} \hat{x}_{0j}(k-1) &= \sum\nolimits_{i=1}^{n} \mu_{i|j}(k-1)\hat{x}_i(k-1), \\ \tilde{p}_{0j}^{opt}(k-1) &= \left(U^T \tilde{P}^{-1}(k-1)U\right)^{-1}. \end{aligned} \tag{47}$$

Step 4 Consider filtering in accordance with mode I, $j \in$ N. Estimation $O_j(k-1)\hat{x}$and variance …are used as input in model j (I, $j \in$ N). Z(k) is used for output of $P_{jj}(k)\hat{x}(k)$, $p_{jj}(k)$Similar functions to filter j are shown as follow:

$$\Lambda_j(k) = \begin{bmatrix} \Lambda_j^1(k) & 0 & \cdots & 0 \\ 0 & \Lambda_j^2(k) & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \Lambda_j^n(kn) \end{bmatrix} \tag{48}$$

$$\begin{aligned} &\Lambda_j^i(k) \\ &= p[z^i(k)|m_k = j, \overset{i}{\hat{x}}_{0j}(k-1), \tilde{p}_{0j}^i(k-1)] \\ &= N[z^i(k), \overset{i}{\hat{z}}_j[k|k-1, \overset{i}{\hat{x}}_{0j}(k-1)], S_j^i[k, \tilde{p}_{0j}^i(k-1)]] \end{aligned} \tag{49}$$

Which below character represents model index and upper character represents mode index.

$\Lambda_j^(i)(k)$probable function related to mode I of filter j and $\tilde{P}_{0j}^(i)(k-1)$ variance of $\tilde{x}_{0j}^(i)(k-1)$

Step 5 Update state probability and combine state estimation and related variance (I, $j \in$ N)

$$\mu_j(k) = \Lambda_j(k)\overline{c}_j c^{-1} \tag{50}$$

In which C is normal fixed matrix.

$$\begin{aligned} c &= \sum_{i=1}^{n} \Lambda_j(k)\overline{c}_j, \\ \hat{x}(k) &= \sum_{i=1}^{n} \mu_j(k)\hat{x}_j(k) \end{aligned} \tag{51}$$

$$P(k) = \sum_{j=1}^{n} \mu_j(k)\left\{ \tilde{P}_{jj}(k) + \left[\hat{x}_j(k) - \hat{x}(k)\right]\left[\hat{x}_j(k) - \hat{x}(k)\right]^T \right\} \tag{52}$$

To solve this problem that Mode Transition Matrix is incorrect and without residence time it can be determined, OMTM-IMM algorithm has been presented based on linear least variance theory. Therefore, Optimal Mode Transition depends on continuous state estimation. Sub-filter relation and made effects by each state. Stimulation results show that accuracy of OMTM-IMM is better than IMM, about 20% computation is incorrect, it can be concluded that OMTM-IMM algorithm is more appropriate for tracking maneuvering target, however complexities of algorithm computations of OMTM-IMM algorithm is a little more than IMM algorithm.

4.2.4 IMM-ARUKF Model: 2017

This paper presents an IMM estimation based adaptive Robust UKF to address this problem. This method combines the merits of the adaptive fading UKF to robust UKF and discard their demerits to inhabit the disturbance of system model uncertainty on the filtering solution and Markov chain is used to transition probability between AFUKF and RUKF. An adaptive fading UKF for the case of process model uncertainty and a robust UKF for the case of measurement model uncertainty are established based on the principle of innovation orthogonality. Subsequently an IMM estimation is developed to fuse the adaptive fading UKF and robust UKF as sub-filters according to the mode probability. The system state estimation is achieved as a probabilities weighted sum of the estimation results from the two sub-filters [11].

The key idea of AFUKF is to include an adaptive fading factor by time in predicted state covariance Matrix, it can prevent the effect of prior knowledge in current state estimation [45].

The key idea of RUKF is to include a time variant robust factor in innovation covariance to control abnormal measurement in current state estimation [46].

It is observed that AFUKF only control process model uncertainty, while, RUKF controls measurement model uncertainty. Therefore, both filters have supplementary features (Fig. 13).

In this method a multiple fusion process issued in which AFUKF and RUKF are implemented as two sub-filters in parallel, each adapts a special system state. A sub-filter for AFUKF is a F U KF process model just in uncertainty condition while RUKF is just about measurement model uncertainty. Transition probabilities between two sub-filter is controlled by a Markov chain. Total state estimation is obtained by combining all state estimations of each sub-filter. Proposed IMM-ARUKF consists of following main steps:

Step 1 Initialization: This step is for starting two fusion processes of sub-filters and some filters. AFUKF and RUKF can be regulated as:

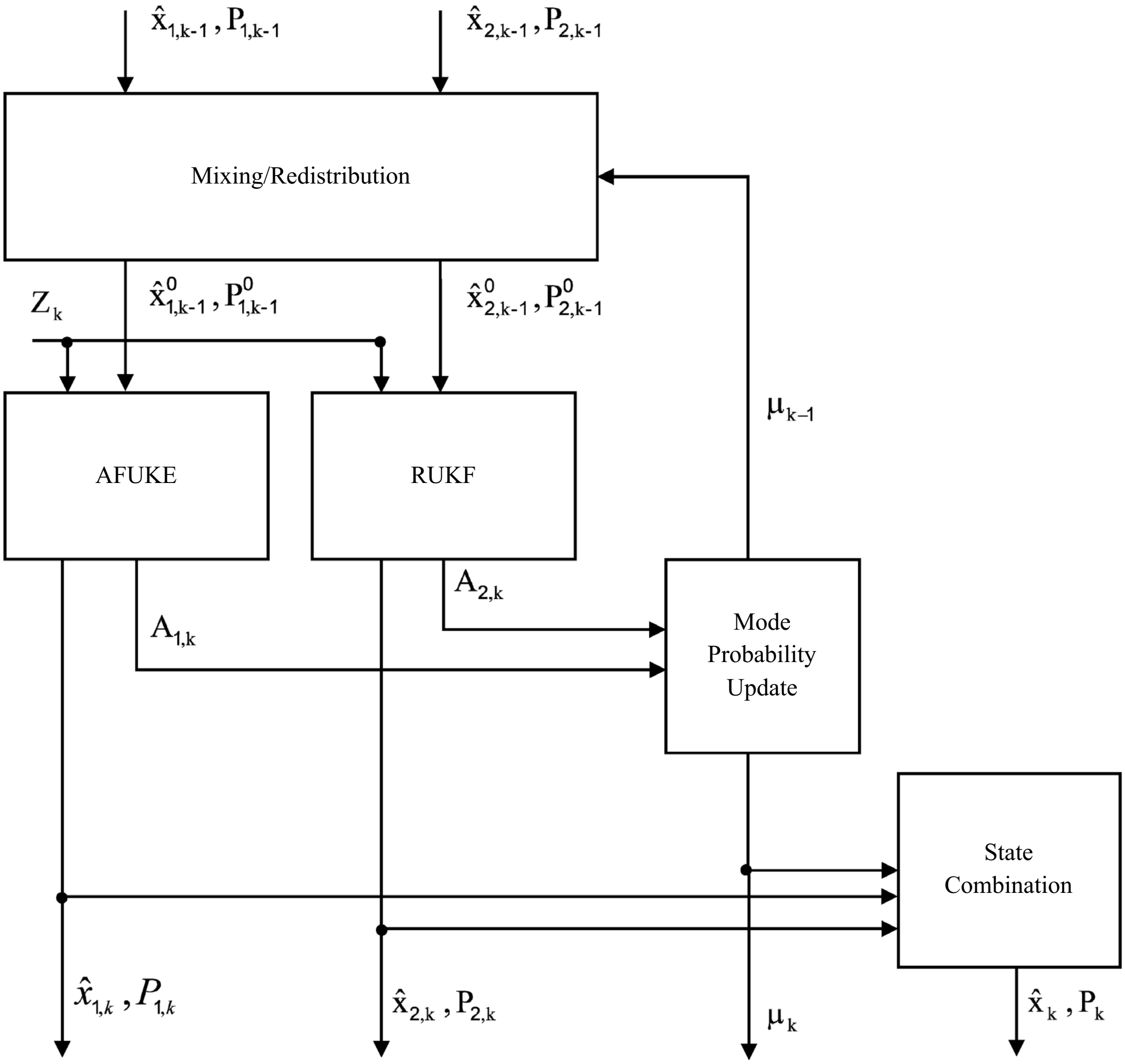


Fig. 13 IMM-ARUKF structure

$$\begin{cases} \hat{x}_{j,0} = E[\hat{x}_{j,0}] \\ P_{j,0} = E[(x_{j,0} - \hat{x}_{j,0})(x_{j,0} - \hat{x}_{j,0})^T]^{(j=1,2)} \end{cases} \tag{53}$$

(ii) Initialization of fusion process of some filters using Markov Transition Matrix and possibility of initial state is performed as follow:

$$M = \begin{bmatrix} m_{11} & m_{11} \\ m_{21} & m_{22} \end{bmatrix} \quad , \qquad \mu_0 = \begin{bmatrix} \mu_{1,0} \\ \mu_{2,0} \end{bmatrix} \tag{54}$$

Step 2 Parallel filter. RUKF and AFUKF are performed in this step in a parallel structure, it is used for estimating related state $X_{j\hat{}k}$. and error covariance matrix P_j, K_k.

A) By replacing predicted state covariance matrix, classic AFUKF is performed by corrected type, as shown below. Other steps are similar to UKF.

$$P^*_{k/k-1} = \lambda_* \{ \sum_{i=0}^{2n} \omega_i (X_{i,k/k} - \hat{x}_{k/k-1})(X_{i,k/k} - \hat{x}_{k/k-1})^T + Q_k \}, \tag{55}$$

Step 3 Fusion of some filters: Based on estimation method of IMM total system state estimation and related error covariance matrix can be calculated as follow:

Table 2 Comparative table of methods

Filter type	Method	Author(s)	Base filter	Compared method	Advantages	Disadvantages	Applications
Single filter	KF	Kalman [1]	–	–	Basic method for prediction, using in linear systems	Not being able to predict in nonlinear systems	Used as a base filter in linear issues, particularly navigation, and control of vehicles
	RKF	Xie et al. [9]	KF	KF	Solving nonlinear systems estimation	High computational cost	For uncertain discrete-time systems with norm-bounded parameter uncertainty
	UKF	Gao et al. [11]	KF	EKF	More accurate in calculation, simpler computational process	Divergent solution when system model is uncertain, need to an accurate system model and accurate noise information	A solution for obtain for upper limit of error covariance for linear and nonlinear systems by using Riccati equation
	CKF	Duan et al. [15]	KF	UKF, EKF	Lower computational cost, better convergence and dimensions	Longer performance time than EKF algorithm	Used as a base filter in nonlinear issues
	PKF	Farahi and Yazdi [16]	KF	KF	Using prior knowledge, solving blocks	High time complexity	It can track a target with abnormal behavior that is suitable for several applications such as traffic and airport control
MMAE	MMAE	Hanlon and Maybeck [20]	KF	KF	Detecting failure in controller/moving sensor	Long time to recognize failure more than standard limit of MMAE, when control input is almost strong	It is used to detect flight control actuator failures
	RMMAE	Xiong et al. [19]	RKF	MMAE	Resistance to model parameter detecting error against MMAE, more stable than RKF	High processing time than EKF	It is implemented for XNAV in the presence of the pulsar angular position error
	SMMAE	Kottath et al. [36]	EKF	MMAE	High accuracy due to many initial models, selecting the most effective model among all models, performance improvement than MMAE	Challenge of initialization of unknown parameter	
	WMMAE	Kottath et al. [39]	EKF	MMAE	Improvement in state estimation considering window size as one of unknown parameters in MMAE framework, creating an adaptive and resistance system to system noise with different window size	Incorrect choice of parameter w may lead to no optimal results	It is able to converge to the optimal filter based on the on-going data stream statistics
IMM	IMM-CKF	Wan et al. [14]	CKF	IMMUKF, IMMEKF	Decrease in estimation errors in comparison by comparative methods		Presented to track the maneuvering target using angular measurements

Table 2 (continued)

Filter type	Method	Author(s)	Base filter	Compared method	Advantages	Disadvantages	Applications
	UKF-MIMM	Gao et al. [43]	UKF	UKF-IMM	Higher accuracy and less computational complexity than EKF-IMM, UKF-IMM	The longest performance time among six mentioned algorithms	Suitable for nonlinear maneuvering target tracking
	UKF-SIMM				Upper velocity than EKF-IMM, UKF-IMM	Longer performance time than EKF algorithms	
	EKF-MIMM		EKF	EKF-IMM	Upper velocity than EKF-IMM	Lower speed than UKF based algorithms	
	EKF-SIMM				The least function time and less computational costs in 4 current methods	More errors in position estimation	
	OMTM-IMM	Sun and Shen [44]	KF	IMM	Accuracy of tracking OMTM-IMM algorithm y than IMM algorithm	Complexity in OMTM-IMM algorithm calculations is more than IMM algorithm	Suitable for the maneuvering target tracking problem
	IMM-ARUKF	Gao et al. [11]	RUKF	UKF, AFUKF, RUKF	Preventing disorder in system measurement model uncertainty in RUKF, improvement in position error	–	Suitable for vehicle navigation
			AFUKF		Controlling filtering disorders because of uncertainty in AFUKF model, flexibility and resistance improvement	–	

$$\hat{x}_k = \sum_{j=1}^{2} \hat{x}_{j,k}\mu_{j,k} \quad (56)$$

$$P_k = \sum_{j=1}^{2} \{P_{j,k} + [\hat{x}_{j,k} - \hat{x}_k][\hat{x}_{j,k} - \hat{x}_k]^T\}\mu_{j,k} \quad (57)$$

Step 4 Repeat step 2, 3 for next sample until processing all samples. The results of experimental stimulation and comparison analysis show that IMM-ARUKF suggestion can prevent system model uncertainty disorders in UKF solution. By comparing the mean square root error of the RMSE algorithm with the classical UKF, AFUKF and RUKF, it is claimed that IMM-ARUKF has an important ability to control filtering disorders due to system model uncertainty. Therefore, it leads to improve filter flexibility and resistance. Situation error improves significantly.

5 Conclusion and Unsolved Problems

One of unsolved problems that can limit its practical applications can be functional applications of KF in LDS, it still has technical problems such as determining initial values of filter, little deviation in model coefficients, systematic deviation of measurement a covariance of model disorder estimation, measurement errors. Initial deviation of filter states ca lead to state deviation of KF for a long time. When partial deviation of model structures make KF states deviated for a long time, it changes to a convergent stable system. It may lead to KF state deviation after being in threshold, by increasing in distance, the affected time on KG state can become longer. When systematic deviation of measurement data makes KF clarity deviant, it never gets convergent in unobservable systems. Whether above-mentioned issues effect on KF estimation and its accuracy or not is a practical problem that is of high importance in KF application and it is inevitable. Most researchers have suggested corresponding solutions but such problems have not been solved yet. Some problems have been ignored, either [47].

Generally, knowing about advantages and disadvantages of a method can help to choose an effective model. Paying attention to required conditions for implementing a system such as time and cost limitations can be considered as an important challenge. For simplifying, advantages and limitations of introduced methods have been shown in Table 2.

As it was mentioned KF is a recursive estimation algorithm. EKF, UKF, and CKF were introduced to solve problems in nonlinear and Gaussian calculations. RKF provides upper limit for covariance of estimation error to control model uncertainty. For a better performance in complicated systems, MM Filters were introduced. By using two or more filters with different models in parallel, more reliable estimations will be obtained. By allocating an estimation to each filter outputs of each filter are calculated. MMAE and IMM are methods that are used mostly for MM estimations. It is supposed that there is no difference between time and model in MMAE algorithm. In IMM, jumps in system model are paid attention to. In MMAE residual of different filters is used to create adaptive weight, while IMM improves MMAE by combining initial conditions of each filter in each stage, calculating delays in lower state estimation.

Generally, it is observed that MMAE is used for airplanes while IMM is used for target tracking problems. Since using these methods has computational costs, the method should be selected carefully based on system factors and computational resources [41].

References

1. Kalman RE (1960) A new approach to linear filtering and prediction problems. J Basic Eng 82(1):35–45
2. Kelly A (1994) A 3D state space formulation of a navigation Kalman filter for autonomous vehicles. Carnegie-Mellon University Robotics Institute, Pittsburgh
3. Reid I, Term H (2001) Estimation II. Lecture notes. University of Oxford, Oxford
4. Bishop G, Welch G (2001) An introduction to the Kalman filter. In: Proceedings of SIGGRAPH, course, 2001, vol 8(27599-23175), p 41
5. Deng Z (2003) Kalman filter and the Wiener filter: modern time series analysis method. Press of Harbin Institute of Technology, Harbin
6. Huang X, Wang Y (2015) The Kalman filter principle and application. Publishing House of Electronics Industry, Beijing
7. Wolpert DM, Ghahramani Z (2000) Computational principles of movement neuroscience. Nat Neurosci 3(11):1212–1217
8. Kang-Hua T, Mei-Ping W, Xiao-Ping H (2007) Multiple model Kalman filtering for MEMS-IMU/GPS integrated navigation. In: 2007 2nd IEEE conference on industrial electronics and applications, 2007. IEEE
9. Xie L, Soh YC, De Souza CE (1994) Robust Kalman filtering for uncertain discrete-time systems. IEEE Trans Autom Control 39(6):1310–1314
10. Julier SJ, Uhlmann JK (1997) New extension of the Kalman filter to nonlinear systems. In: Signal processing, sensor fusion, and target recognition VI, 1997. International Society for Optics and Photonics
11. Gao B et al (2017) Interacting multiple model estimation-based adaptive robust unscented Kalman filter. Int J Control Autom Syst 15(5):2013–2025
12. Gustafsson F, Hendeby G (2011) Some relations between extended and unscented Kalman filters. IEEE Trans Signal Process 60(2):545–555
13. Arasaratnam I, Haykin S (2009) Cubature Kalman filters. IEEE Trans Autom Control 54(6):1254–1269
14. Wan M, Li P, Li T (2010) Tracking maneuvering target with angle-only measurements using IMM algorithm based on CKF.

In: 2010 International conference on communications and mobile computing, 2010. IEEE
15. Duan J et al (2016) Square root cubature Kalman filter-Kalman filter algorithm for intelligent vehicle position estimate. Procedia Eng 137:267–276
16. Farahi F, Yazdi HS (2020) Probabilistic Kalman filter for moving object tracking. Signal Process Image Commun 82:115751
17. Forney GD (1973) The Viterbi algorithm. Proc IEEE 61(3):268–278
18. Forney GD Jr (2005) The Viterbi algorithm: a personal history. arXiv preprint cs/0504020
19. Xiong K, Wei C, Liu L (2015) Robust multiple model adaptive estimation for spacecraft autonomous navigation. Aerosp Sci Technol 42:249–258
20. Hanlon PD, Maybeck PS (2000) Multiple-model adaptive estimation using a residual correlation Kalman filter bank. IEEE Trans Aerosp Electron Syst 36(2):393–406
21. Seah CE, Hwang I (2009) State estimation for stochastic linear hybrid systems with continuous-state-dependent transitions: an IMM approach. IEEE Trans Aerosp Electron Syst 45(1):376–392
22. Li XR, Jilkov VP (2005) Survey of maneuvering target tracking. Part V. Multiple-model methods. IEEE Trans Aerosp Electron Syst 41(4):1255–1321
23. Izadian A, Famouri P (2010) Fault diagnosis of MEMS lateral comb resonators using multiple-model adaptive estimators. IEEE Trans Control Syst Technol 18(5):1233–1240
24. Yang R et al (2011) RF emitter geolocation using amplitude comparison with auto-calibrated relative antenna gains. IEEE Trans Aerosp Electron Syst 47(3):2098–2110
25. Magill D (1965) Optimal adaptive estimation of sampled stochastic processes. IEEE Trans Autom Control 10(4):434–439
26. Blom HA, Bar-Shalom Y (1988) The interacting multiple model algorithm for systems with Markovian switching coefficients. IEEE Trans Autom Control 33(8):780–783
27. Li X-R, Bar-Shalom Y (1996) Multiple-model estimation with variable structure. IEEE Trans Autom Control 41(4):478–493
28. Chang C-B, Athans M (1977) Hypothesis testing and state estimation for discrete systems with finite-valued switching parameters. Massachusetts Institute of Technology Cambridge Electronic Systems Lab
29. Chang C-B, Athans M (1978) State estimation for discrete systems with switching parameters. IEEE Trans Aerosp Electron Syst 3:418–425
30. Hawkes RM, Moore JB (1976) Performance bounds for adaptive estimation. Proc IEEE 64(8):1143–1150
31. Lainiotis DG (1976) Partitioning: A unifying framework for adaptive systems, I: Estimation. Proc IEEE 64(8):1126–1143
32. Lashlee RW, Maybeck PS (1988) Space structure control using moving bank multiple model adaptive estimation. In: Proceedings of the 27th IEEE conference on decision and control, 1988. IEEE
33. Morales-Menéndez R (n.d.) Real-time monitoring and diagnosis in dynamic systems using particle filtering methods. Edición Única
34. Kay SM (1988) Modern spectral estimation: theory and application. Pearson Education India, Englewood Cliffs
35. Xiong K, Wei C, Liu L (2012) Robust Kalman filtering for discrete-time nonlinear systems with parameter uncertainties. Aerosp Sci Technol 18(1):15–24
36. Kottath R et al (2015) Improving multiple model adaptive estimation by filter stripping. In: 2015 IEEE recent advances in intelligent computational systems (RAICS), 2015. IEEE
37. Hide C, Moore T, Smith M (2004) Adaptive Kalman filtering algorithms for integrating GPS and low cost INS. In: PLANS 2004. Position location and navigation symposium. IEEE Cat. No. 04CH37556, 2004. IEEE
38. Mohamed A, Schwarz K (1999) Adaptive Kalman filtering for INS/GPS. J Geod 73(4):193–203
39. Kottath R et al (2016) Window based multiple model adaptive estimation for navigational framework. Aerosp Sci Technol 50:88–95
40. Hwang I, Balakrishnan H, Tomlin C (2003) Performance analysis of hybrid estimation algorithms. In: 42nd IEEE international conference on decision and control (IEEE Cat. No. 03CH37475), 2003. IEEE
41. Akca A, Efe MÖ (2019) Multiple model Kalman and Particle filters and applications: a survey. IFAC-PapersOnLine 52(3):73–78
42. Orguner U (2013) EE793 target tracking. Lecture notes
43. Gao L et al (2012) Improved IMM algorithm for nonlinear maneuvering target tracking. Procedia Eng 29:4117–4123
44. Sun L, Shen C (2014) An improved interacting multiple model algorithm used in aircraft tracking. Math Probl Eng 20:14. https://doi.org/10.1155/2014/813654
45. Hu G et al (2015) Modified strong tracking unscented Kalman filter for nonlinear state estimation with process model uncertainty. Int J Adapt Control Signal Process 29(12):1561–1577
46. Jwo D-J, Weng T-P (2008) An adaptive sensor fusion method with applications in integrated navigation. IFAC Proc Vol 41(2):9002–9007
47. Song H, Hu S (2019) Open problems in applications of the Kalman filtering algorithm. In: 2019 International conference on mathematics, big data analysis and simulation and modelling (MBDASM 2019), 2019. Atlantis Press